

CME 241 Project Proposal

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Project Title: Multi-Capital Decision-Making Under Energy Constraints

1. Problem Statement

This project addresses a very general, idealized life optimization problem faced by early-career professionals who are also competitive athletes. Specifically, I focus on a young professional (myself; age 22-30) moving to the Main Line suburbs of Philadelphia to start a new job in Malvern while maintaining a serious weightlifting training program. The challenge is optimizing sequential decisions about housing, training, investments, and consumption to maximize lifetime utility under constraints of limited financial capital, time, finite energy capacity, and stochastic work demands.

Unlike traditional personal finance optimization problems that focus solely on wealth maximization, this formulation recognizes that:

1. Energy is a scarce resource: Physical and mental energy are depleted by work, commuting, and training, and must be actively recovered
2. Money does not necessarily correspond to life quality: Sometimes financially "suboptimal" choices (expensive apartment near gym) yield better life outcomes
3. Training provides direct utility: Athletic performance and training quality are primary sources of life satisfaction, not just costs to be minimized

This problem involves allocating limited resources across:

- Wealth Building: Investments, emergency fund, retirement savings
 - Location Quality: Housing cost vs. commute time vs. neighborhood amenities (nightlife, cafes, walkability)
 - Athletic Performance: Training intensity, coaching, nutrition, recovery vs. injury risk
 - Life Enjoyment: Discretionary consumption, social activities, experiences
 - Energy Management: Balancing training capacity with stochastic work demands and commute time
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2. Practical Relevance

This problem is relevant for several reasons:

- I am currently facing this exact decision problem as I prepare to move to the Main Line suburbs for work in Malvern
 - Many young professionals face similar tradeoffs between career, fitness, location, and lifestyle
 - Most finance RL focuses on pure portfolio optimization; few incorporate energy constraints and multi-capital optimization
 - Very complicated, idealized problem demonstrates curse of dimensionality, curse of modeling, and the value of RL over classical DP. The problem will need to be greatly simplified in each stage to be tractable.
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3. Version 1: Ideal Version

MDP Specification: $\mathcal{M}_1 = (S_1, A_1, P_1, R_1, \gamma)$

State Space S_1

Continuous, 47-dimensional state space capturing rich detail:

Financial State (7 vars):

- $W_t \in [0, \infty)$: Liquid wealth (checking/savings, \$)
- $E_t \in [0, \infty)$: Emergency fund (\$)
- $p_t^{stocks} \in [0, \infty)$: Stock holdings (\$)
- $p_t^{bonds} \in [0, \infty)$: Bond holdings (\$)
- $r_t^{401k} \in [0, \infty)$: 401k balance (\$)
- $r_t^{IRA} \in [0, \infty)$: IRA balance (\$)
- $D_t \in [0, \infty)$: Outstanding debt (\$)
- $S_t \in [60000, 200000]$: Annual salary (\$)
- $B_t \in [0, 50000]$: Expected annual bonus (\$)

Housing/Location State (17 vars):

- $x_t \in [39.95, 40.10]$: Latitude coordinate in Main Line suburbs (Malvern area)
- $y_t \in [-75.60, -75.35]$: Longitude coordinate
- $h_t^{rent} \in [1000, 3500]$: Monthly rent (\$)
- $h_t^{size} \in [400, 1500]$: Apartment size (sq ft)
- $h_t^{bedrooms} \in \{0, 1, 2\}$: Number of bedrooms (studio, 1br, 2br)
- $h_t^{quality} \in [0, 100]$: Overall quality index (fixtures, finishes, maintenance)
- $h_t^{noise} \in [0, 100]$: Noise level (0=quiet, 100=very loud)
- $h_t^{laundry} \in \{0, 1\}$: In-unit laundry (binary)
- $h_t^{storage} \in \{0, 1\}$: Bike/ski storage available (binary)
- $n_t^{safety} \in [0, 100]$: Neighborhood safety score
- $n_t^{walkability} \in [0, 100]$: Walk score
- $n_t^{nightlife} \in [0, 100]$: Nightlife density/quality
- $n_t^{cafes} \in [0, 100]$: Coffee shop density/quality
- $n_t^{parks} \in [0, 100]$: Access to green space
- $c_t^{work} \in [5, 90]$: Commute time to work (minutes)
- $c_t^{gym} \in [5, 60]$: Distance to gym (minutes)
- $c_t^{grocery} \in [2, 30]$: Distance to grocery store (minutes)

Work State (2 vars):

- $W_t^{intensity} \in [0, 100]$: Work intensity/demand (stochastic, exogenous)
- $j_t^{performance} \in [0, 100]$: Job performance score

Energy/Capacity State (7 vars):

- $\Phi_t \in [0, 100]$: Physical energy level
- $\Psi_t \in [0, 100]$: Mental energy level
- $\Theta_t \in [0, 100]$: Accumulated training fatigue
- $Q_t \in [0, 100]$: Sleep quality index
- $N_t \in [0, 100]$: Nutrition quality index
- $\Gamma_t \in [0, 100]$: Injury severity (0=healthy, 100=severely injured)
- $\Sigma_t \in [0, 100]$: Stress level

Training/Performance State (7 vars):

- $m_t^{snatch} \in [40, 140]$: Snatch 1RM (kg)
- $m_t^{C\&J} \in [50, 180]$: Clean & Jerk 1RM (kg)
- $m_t^{squat} \in [80, 250]$: Back squat 1RM (kg)
- $v_t^{mileage} \in [0, 100]$: Recent running/walking mileage (miles/week, 8-week EWMA)
- $v_t^{load} \in [0, 500]$: Recent training load (tons/week, 8-week EWMA)
- $v_t^{intensity} \in [100, 180]$: Average HR per session (bpm, 8-week EWMA)
- $F_t \in [0, 26]$: Time until next competition (biweeks)

Market/Economic State (6 vars):

- $\mu_t^{stock} \in [-0.20, 0.30]$: Expected stock return (annualized)
- $\mu_t^{bond} \in [-0.05, 0.10]$: Expected bond return (annualized)
- $r_t^{rf} \in [0, 0.06]$: Risk-free rate (annualized)
- $\pi_t \in [-0.02, 0.08]$: Inflation rate (annualized)
- $X_t^{labor} \in [0, 100]$: Labor market strength for early-career professionals

Time (1 var):

- $\tau_t \in \{0, 1, \dots, 77\}$: Biweek (0 = start, 77 = end of 3-year horizon)

Total: 47 state dimensions

Action Space A_1

Continuous and discrete, 21-dimensional action space:

Housing (4 actions, yearly decisions):

- $a_t^{move} \in \{0, 1\}$: Whether to move apartments
- $a_t^{location} = (x_t^{new}, y_t^{new}) \in \mathbb{R}^2$: New location coordinates (if moving)
- $a_t^{rent_budget} \in [1000, 3500]$: Monthly rent budget (if moving)

- $a_t^{lease} \in \{6, 12, 18\}$: Lease length in months (if moving)

Financial (7 actions, biweekly decisions):

- $a_t^{invest} \in [0, S_t/26]$: Investment contribution this biweek
- $\alpha_t^{stocks} \in [0, 1]$: Portfolio allocation to stocks (rest to bonds)
- $a_t^{401k} \in [0, \min(0.20 \cdot S_t/26, 23000/26)]$: 401k contribution (pre-tax, employer match)
- $a_t^{IRA} \in [0, 7000/26]$: IRA contribution (post-tax)
- $a_t^{emergency} \in [0, S_t/26]$: Emergency fund contribution
- $a_t^{debt} \in [0, W_t]$: Debt repayment amount
- $a_t^{rebalance} \in \{0, 1\}$: Rebalance portfolio to target allocation

Training (5 actions, biweekly decisions):

- $a_t^{training_intensity} \in [0, 100]$: Training program intensity (0=rest, 100=max)
- $a_t^{coaching} \in \{0, 1, 2\}$: Coaching tier (0=self, 1=online, 2=in-person)
- $a_t^{nutrition} \in [50, 500]$: Nutrition budget per biweek (\$)
- $a_t^{recovery} \in [0, 200]$: Recovery investment (massage, PT, etc., \$ per biweek)
- $a_t^{compete} \in \{0, 1\}$: Compete in upcoming competition (binary)

Consumption/Lifestyle (5 actions, biweekly decisions):

- $a_t^{dining} \in [0, 500]$: Dining out budget per biweek (\$)
- $a_t^{social} \in [0, 300]$: Social activities budget (bars, events, \$ per biweek)
- $a_t^{hobbies} \in [0, 200]$: Hobbies/entertainment budget (\$ per biweek)
- $a_t^{travel} \in \{0, 1\}$: Take a trip this quarter (binary, avg \$1500/trip)

Total: 21 action dimensions (4 housing + 7 financial + 5 training + 5 consumption)

Transition Dynamics $P_1(s_{t+1} | s_t, a_t)$

Financial Transitions:

$$W_{t+1} = W_t + \frac{S_t}{26} - \frac{h_t^{rent}}{26} - a_t^{invest} - a_t^{401k} - a_t^{IRA} - a_t^{emergency} - a_t^{debt} - a_t^{dining} - a_t^{social} - a_t^{hobbies} - 1500 \cdot a_t^{travel} + \epsilon_t^W$$

where $\epsilon_t^W \sim \mathcal{N}(0, 100^2)$ captures unexpected expenses/income.

$$E_{t+1} = E_t + a_t^{emergency}$$

$$p_{t+1}^{stocks} = p_t^{stocks} + \alpha_t^{stocks} \cdot a_t^{invest} + r_{t+1}^{stocks} \cdot p_t^{stocks}$$

$$p_{t+1}^{bonds} = p_t^{bonds} + (1 - \alpha_t^{stocks}) \cdot a_t^{invest} + r_{t+1}^{bonds} \cdot p_t^{bonds}$$

where $r_{t+1}^{stocks} \sim \mathcal{N}(\mu_t^{stock}/26, (0.18/\sqrt{26})^2)$ and $r_{t+1}^{bonds} \sim \mathcal{N}(\mu_t^{bond}/26, (0.06/\sqrt{26})^2)$ are biweekly returns.

$$r_{t+1}^{401k} = r_t^{401k} + a_t^{401k} \cdot (1 + \text{match_rate}) + r_{t+1}^{port} \cdot r_t^{401k}$$

$$r_{t+1}^{IRA} = r_t^{IRA} + a_t^{IRA} + r_{t+1}^{port} \cdot r_t^{IRA}$$

where $r_{t+1}^{port} = 0.7 \cdot r_{t+1}^{stocks} + 0.3 \cdot r_{t+1}^{bonds}$ (target allocation).

$$D_{t+1} = \max(0, D_t \cdot (1 + r_t^{debt}/26) - a_t^{debt})$$

Salary Transitions:

$$S_{t+1} = \begin{cases} S_t \cdot (1 + 0.15 + \epsilon_t^{promo}) & \text{if promoted} \\ S_t \cdot (1 + 0.03 + \epsilon_t^{raise}) & \text{annual raise (every 26 biweeks)} \\ S_t & \text{otherwise} \end{cases}$$

where $P(\text{promoted}) = \text{logistic}(-3 + 0.03 \cdot j_t^{performance} + 0.02 \cdot \tau_t)$ and $\epsilon_t^{promo}, \epsilon_t^{raise} \sim \mathcal{N}(0, 0.02^2)$.

Housing Transitions:

If $a_t^{move} = 1$:

- $(x_{t+1}, y_{t+1}) = a_t^{location}$
- $h_{t+1}^{rent} = a_t^{rent_budget}$
- Other housing characteristics $(h_{t+1}^{size}, h_{t+1}^{quality}, \dots)$ sampled from distribution conditioned on location and rent:

$$P(H_{t+1} | x_{t+1}, y_{t+1}, h_{t+1}^{rent})$$

- Neighborhood characteristics are deterministic functions of location:

$$n_{t+1}^{safety} = f_{safety}(x_{t+1}, y_{t+1}), \quad n_{t+1}^{walkability} = f_{walk}(x_{t+1}, y_{t+1}), \dots$$

- Commute times are deterministic functions of location and work/gym locations:

$$c_{t+1}^{work} = \text{distance}((x_{t+1}, y_{t+1}), (x_{work}, y_{work})), \dots$$

If $a_t^{move} = 0$:

- All housing/location variables remain unchanged
- $h_{t+1}^{rent} = h_t^{rent} \cdot (1 + 0.03 + \epsilon_t^{rent})$ (annual rent increase)

Work Intensity Transition (exogenous, stochastic):

$$W_{t+1}^{intensity} = 0.7 \cdot W_t^{intensity} + 0.3 \cdot 50 + \epsilon_t^{work}$$

where $\epsilon_t^{work} \sim \mathcal{N}(0, 15^2)$ (mean-reverting AR(1) around 50).

Job Performance Transition:

$$j_{t+1}^{performance} = \text{clip}(0.6 \cdot j_t^{performance} + 0.25 \cdot \Psi_t - 0.15 \cdot \Sigma_t + 0.1 \cdot \min(\tau_t, 52) + \epsilon_t^{perf}, 0, 100)$$

where $\epsilon_t^{perf} \sim \mathcal{N}(0, 8^2)$. Performance depends on mental energy, stress, tenure, and random factors.

Energy Transitions:

$$\begin{aligned} \Phi_{t+1} = & \text{clip}(\Phi_t - 0.4 \cdot W_t^{intensity}/100 - 0.35 \cdot v_t^{load}/500 - 0.02 \cdot (c_t^{work} + c_t^{gym}) \\ & + 25 \cdot 1_{Q_t > 80} + 15 \cdot 1_{N_t > 75} - 10 \cdot \Gamma_t/100 + \epsilon_t^{\Phi}, 0, 100) \end{aligned}$$

where $\epsilon_t^\Phi \sim \mathcal{N}(0, 5^2)$.

$$\begin{aligned}\Psi_{t+1} = & \text{clip}(\Psi_t - 0.5 \cdot W_t^{\text{intensity}}/100 - 0.3 \cdot \Sigma_t - 0.01 \cdot (c_t^{\text{work}} + c_t^{\text{gym}}) \\ & + 20 \cdot 1_{Q_t > 80} + 0.15 \cdot n_t^{\text{walkability}} + \epsilon_t^\Psi, 0, 100)\end{aligned}$$

where $\epsilon_t^\Psi \sim \mathcal{N}(0, 6^2)$.

$$\Theta_{t+1} = \text{clip}(\Theta_t + v_t^{\text{load}}/100 + v_t^{\text{intensity}}/10 - \text{recovery}(\mathbf{N}_t, Q_t, a_t^{\text{recovery}}) + \epsilon_t^\Theta, 0, 100)$$

where recovery function: $\text{recovery}(\mathbf{N}, Q, r) = 0.3 \cdot \mathbf{N}/100 + 0.4 \cdot Q/100 + 0.01 \cdot r$.

$$Q_{t+1} = \text{clip}(60 - 0.3 \cdot h_t^{\text{noise}} - 0.2 \cdot \Sigma_t - 0.15 \cdot \Theta_t + 0.1 \cdot n_t^{\text{parks}} + \epsilon_t^Q, 0, 100)$$

$$\mathbf{N}_{t+1} = \text{clip}(40 + 0.12 \cdot a_t^{\text{nutrition}} - 0.2 \cdot W_t^{\text{intensity}}/100 + \epsilon_t^\mathbf{N}, 0, 100)$$

$$\Gamma_{t+1} = \max(0, \Gamma_t - 5 + 20 \cdot 1_{\text{injury event}} + \epsilon_t^\Gamma)$$

where $P(\text{injury event}) = \text{logistic}(-6 + 0.05 \cdot \Theta_t + 0.02 \cdot v_t^{\text{load}} - 0.03 \cdot a_t^{\text{recovery}})$.

$$\begin{aligned}\Sigma_{t+1} = & \text{clip}(0.6 \cdot \Sigma_t + 0.3 \cdot W_t^{\text{intensity}}/100 + 20 \cdot 1_{W_t < 5000} + 15 \cdot 1_{E_t < 3 \cdot h_t^{\text{rent}}} \\ & - 0.1 \cdot n_t^{\text{nightlife}} - 0.05 \cdot n_t^{\text{cafes}} + \epsilon_t^\Sigma, 0, 100)\end{aligned}$$

Training Volume Transitions (EWMA of recent 8 weeks):

$$v_{t+1}^{\text{mileage}} = 0.85 \cdot v_t^{\text{mileage}} + 0.15 \cdot \text{mileage}(a_t^{\text{training_intensity}}, \Phi_t)$$

$$v_{t+1}^{\text{load}} = 0.85 \cdot v_t^{\text{load}} + 0.15 \cdot \text{load}(a_t^{\text{training_intensity}}, \Phi_t, a_t^{\text{coaching}})$$

$$v_{t+1}^{\text{intensity}} = 0.85 \cdot v_t^{\text{intensity}} + 0.15 \cdot \text{HR}(a_t^{\text{training_intensity}}, \Phi_t)$$

where current training depends on intensity choice and available energy.

Strength Transitions:

$$m_{t+1}^{\text{snatch}} = m_t^{\text{snatch}} + \text{adaptation}(v_t^{\text{load}}, a_t^{\text{coaching}}, \Theta_t, \Phi_t, \mathbf{N}_t, \Gamma_t) + \epsilon_t^{\text{snatch}}$$

where $\text{adaptation}(\dots) = (0.005 \cdot v_t^{\text{load}} + 0.2 \cdot (a_t^{\text{coaching}} - 0.5)) \cdot (1 - \Theta_t/100) \cdot (\Phi_t/100)^{0.5} \cdot (\mathbf{N}_t/100) \cdot (1 - \Gamma_t/100)$.

Similar expressions for $m_{t+1}^{C\&J}$ and m_{t+1}^{squat} .

Market Transitions:

$$\mu_{t+1}^{\text{stock}} = 0.8 \cdot \mu_t^{\text{stock}} + 0.2 \cdot 0.08 + \epsilon_t^{\mu_{\text{stock}}}$$

$$\mu_{t+1}^{\text{bond}} = 0.85 \cdot \mu_t^{\text{bond}} + 0.15 \cdot 0.03 + \epsilon_t^{\mu_{\text{bond}}}$$

$$r_{t+1}^{rf} = 0.9 \cdot r_t^{rf} + 0.1 \cdot 0.03 + \epsilon_t^{rf}$$

$$\pi_{t+1} = 0.7 \cdot \pi_t + 0.3 \cdot 0.02 + \epsilon_t^\pi$$

All $\epsilon_t^{\text{market}} \sim \mathcal{N}(0, \sigma_{\text{market}}^2)$ with appropriate volatilities.

Labor Market Transition:

$$X_{t+1}^{\text{labor}} = 0.9 \cdot X_t^{\text{labor}} + 0.1 \cdot 50 + \epsilon_t^{\text{labor}}$$

where $\epsilon_t^{labor} \sim \mathcal{N}(0, 10^2)$ (mean-reverting AR(1) around 50).

Time Transition: $\tau_{t+1} = \tau_t + 1$

Competition Schedule: $F_{t+1} = \max(0, F_t - 1)$ (counts down to competition)

Reward Function $R_1(s_t, a_t)$

Multi-objective utility function combining 7 components:

$$R_1(s_t, a_t) = \alpha_1 U^{training}(s_t, a_t) + \alpha_2 U^{consumption}(a_t) + \alpha_3 U^{housing}(s_t) \\ + \alpha_4 U^{security}(s_t) + \alpha_5 U^{health}(s_t) + \alpha_6 U^{career}(s_t) - \alpha_7 C^{stress}(s_t)$$

1. Training Utility ($\alpha_1 = 8$):

$$U^{training}(s_t, a_t) = 0.4 \cdot \frac{m_t^{snatch} + m_t^{C\&J} + 0.5 \cdot m_t^{squat}}{200} + 0.3 \cdot \frac{\Phi_t}{100} \cdot \frac{v_t^{load}}{500} \\ + 0.2 \cdot (1 - \frac{\Theta_t}{100}) + 0.1 \cdot (a_t^{coaching} + 1) + 50 \cdot a_t^{compete}$$

Combines strength level (40%), training quality (30%), recovery state (20%), coaching quality (10%), and competition participation bonus.

2. Consumption Utility ($\alpha_2 = 2$):

$$U^{consumption}(a_t) = 10 \cdot \log(1 + a_t^{dining}/100) + 8 \cdot \log(1 + a_t^{social}/100) \\ + 5 \cdot \log(1 + a_t^{hobbies}/100) + 30 \cdot a_t^{travel}$$

Logarithmic utility from discretionary spending (diminishing returns) and travel bonus.

3. Housing Utility ($\alpha_3 = 3$):

$$U^{housing}(s_t) = 0.15 \cdot n_t^{nightlife} + 0.15 \cdot n_t^{cafes} + 0.20 \cdot n_t^{walkability} \\ + 0.10 \cdot n_t^{parks} + 0.15 \cdot h_t^{quality} - 0.10 \cdot h_t^{noise} \\ + 15 \cdot h_t^{laundry} + 10 \cdot h_t^{storage} - 0.05 \cdot (c_t^{work} + c_t^{gym})$$

Values neighborhood amenities, housing quality, and penalizes commute time.

4. Financial Security Utility ($\alpha_4 = 2.5$):

$$U^{security}(s_t) = 15 \cdot \sqrt{\frac{W_t}{10000}} + 10 \cdot \sqrt{\frac{E_t}{10000}} \\ + 5 \cdot \sqrt{\frac{p_t^{stocks} + p_t^{bonds}}{10000}} + 3 \cdot \sqrt{\frac{r_t^{401k} + r_t^{IRA}}{10000}} - 20 \cdot \sqrt{\frac{D_t}{1000}}$$

CRRA utility (diminishing returns) from wealth, emergency fund, investments, retirement savings; debt penalty.

5. Health Utility ($\alpha_5 = 2$):

$$U^{health}(s_t) = 0.3 \cdot Q_t + 0.25 \cdot N_t + 0.25 \cdot \Phi_t + 0.20 \cdot \Psi_t - 30 \cdot \frac{\Gamma_t}{100}$$

Values sleep, nutrition, physical/mental energy; severe injury penalty.

6. Career Utility ($\alpha_6 = 1.5$):

$$U^{career}(s_t) = 0.5 \cdot \frac{S_t - 60000}{140000} + 0.3 \cdot j_t^{performance} + 0.2 \cdot X_t^{labor}$$

Values salary growth, job performance, and favorable labor market conditions.

7. Stress Cost ($\alpha_7 = 3$):

$$C^{stress}(s_t) = \left(\frac{\Sigma_t}{100}\right)^2 + 40 \cdot 1_{W_t < 3000} \cdot 1_{E_t < 6 \cdot h_t^{rent}}$$

Quadratic stress disutility and acute financial insecurity penalty (low liquid wealth + insufficient emergency fund).

Total Reward Range: Approximately [-50, 250] per biweek, with training and housing quality dominating for high-performing policies.

Discount Factor

$\gamma = 0.995$ (biweekly discounting over 3-year horizon: 78 time steps)

4. Realistic RL Version (Phase 3)

MDP Specification: $\mathcal{M}_2 = (S_2, A_2, P_2, R_2, \gamma)$

Goal: Solvable with modern deep RL (PPO, SAC) by end of course, while capturing realistic complexity.

State Space S_2

Mixed discrete/continuous, 21-dimensional state space:

$$s_t = (W_t, E_t, I_t, R_t, L_t, h_t^{rent}, h_t^{commute}, W_t^{intensity}, \Phi_t, \Psi_t, \Theta_t, \Sigma_t, m_t^{snatch}, m_t^{C\&J}, V_t, F_t, S_t, j_t^{perf}, M_t^{market}, Q_t, \tau_t)$$

Financial State (4 vars):

- $W_t \in [0, 200000]$: Liquid wealth (continuous, \$)
- $E_t \in \{0, 1, 2\}$: Emergency fund level (0=none, 1=partial, 2=full ≥ 6 mo expenses)
- $I_t \in [0, 500000]$: Investment portfolio value (continuous, aggregate stocks/bonds, \$)
- $R_t \in [0, 300000]$: Retirement savings (continuous, aggregate 401k/IRA, \$)

Housing/Location State (3 vars):

- $L_t \in \{0, 1, \dots, 7\}$: Neighborhood ID (8 discrete zones: Malvern/Paoli, Wayne/Berwyn, Ardmore, West Chester, King of Prussia, Radnor/Villanova, Devon/Frazer, Conshohocken)
- $h_t^{rent} \in [1000, 3500]$: Monthly rent (continuous, \$)
- $h_t^{commute} \in [10, 120]$: Total commute time to work + gym (continuous, minutes)

Work State (1 var):

- $W_t^{intensity} \in [0, 100]$: Work intensity/demand this period (stochastic, exogenous)

Energy/Capacity State (5 vars):

- $\Phi_t \in [0, 100]$: Physical energy (continuous)
- $\Psi_t \in [0, 100]$: Mental energy (continuous)
- $\Theta_t \in [0, 100]$: Accumulated training fatigue (continuous)
- $\Sigma_t \in [0, 100]$: Stress level (continuous)
- $Q_t \in [0, 100]$: Sleep quality (continuous)

Training/Performance State (4 vars):

- $m_t^{snatch} \in [40, 140]$: Snatch 1RM (continuous, kg)
- $m_t^{C\&J} \in [50, 180]$: Clean & Jerk 1RM (continuous, kg)
- $V_t \in [0, 500]$: Aggregate training load index (continuous, tons/week equivalent)
- $F_t \in \{0, 1\}$: Competition this quarter (binary)

Career State (2 vars):

- $S_t \in [60000, 200000]$: Annual salary (continuous, \$)
- $j_t^{perf} \in [0, 100]$: Job performance score (continuous, simplified from seniority levels)

Market State (1 var):

- $M_t^{market} \in \{0, 1, 2\}$: Market regime (0=bear, 1=neutral, 2=bull)

Time (1 var):

- $\tau_t \in \{0, 1, \dots, 77\}$: Biweek (3-year horizon)

Total: 21 state dimensions (mixed discrete/continuous)

Action Space A_2

Mixed discrete/continuous, 9-dimensional action space:

$$a_t = (a_t^{housing}, a_t^{invest}, a_t^{allocation}, a_t^{emergency}, a_t^{intensity}, a_t^{coaching}, a_t^{nutrition}, a_t^{discretionary}, a_t^{compete})$$

Housing (1 action, yearly decision):

- $a_t^{housing}$: If moving, choose from 8 suburban zones + rent tier (discrete set, ~24 options)

Financial (3 actions, biweekly):

- $a_t^{invest} \in [0, S_t]$: Investment amount (continuous)
- $a_t^{allocation} \in [0, 100]$: % in stocks (continuous, rest in bonds)
- $a_t^{emergency} \in [0, S_t]$: Emergency fund contribution (continuous)

Training (3 actions, biweekly):

- $a_t^{intensity} \in \{0, 1, 2\}$: Training intensity (maintenance/moderate/aggressive)
- $a_t^{coaching} \in \{0, 1\}$: Online coaching (binary)

- $a_t^{nutrition} \in \{0, 1, 2\}$: Nutrition budget tier (basic/moderate/premium)

Consumption (2 actions, biweekly):

- $a_t^{discretionary} \in [0, S_t]$: Discretionary spending budget (continuous)
- $a_t^{compete} \in \{0, 1\}$: Compete this quarter (binary)

Total: 9 action dimensions (mixed discrete/continuous)

Transition Dynamics $P_2(s_{t+1} | s_t, a_t)$

Financial Transitions:

$$W_{t+1} = W_t + \frac{S_t}{26} - \frac{\text{rent}(L_t)}{26} - a_t^{discretionary} - a_t^{invest} - a_t^{emergency} + \epsilon_t^W$$

$$I_{t+1} = I_t + a_t^{invest} + r_{t+1}(M_t^{market}, a_t^{allocation}) \cdot I_t$$

where $r_{t+1} \sim \mathcal{N}(\mu(M_t^{market}, a_t^{allocation}), \sigma^2)$ (stochastic market returns)

Work Intensity Transition (stochastic, exogenous):

$$W_{t+1}^{intensity} = 0.7 \cdot W_t^{intensity} + 0.3 \cdot 50 + \epsilon_t^{work}$$

where $\epsilon_t^{work} \sim \mathcal{N}(0, 15^2)$ (mean-reverting AR(1) process around 50)

Energy Transitions:

$$\Phi_{t+1} = \text{clip}(\Phi_t - 0.5 \cdot W_t^{intensity}/100 - 0.3 \cdot V_t - 0.01 \cdot h_t^{commute} + 20 \cdot 1_{Q_t > 80} + \epsilon_t^\Phi, 0, 100)$$

$$\Psi_{t+1} = \text{clip}(\Psi_t - 0.6 \cdot W_t^{intensity}/100 - 0.5 \cdot \Sigma_t + 0.2 \cdot Q_t + \epsilon_t^\Psi, 0, 100)$$

$$\Theta_{t+1} = \text{clip}(\Theta_t + V_t - \text{recovery}(a_t^{nutrition}, Q_t), 0, 100)$$

Note: Energy is **depleted by stochastic work intensity** (not chosen), creating periods of high/low energy availability.

Training Transitions:

$$m_{t+1}^{snatch} = m_t^{snatch} + \text{adaptation}(V_t, a_t^{coaching}, \Theta_t, \Phi_t, a_t^{nutrition}) + \epsilon_t^M$$

$$V_t = f(a_t^{intensity}, \Phi_t, a_t^{coaching})$$

(training volume depends on chosen intensity, available energy, coaching)

Career Transitions:

$$S_{t+1} \sim \begin{cases} S_t \cdot (1 + 0.15 + \epsilon_t^S) & \text{if promoted} \\ S_t \cdot (1 + 0.03 + \epsilon_t^S) & \text{otherwise (annual raise)} \end{cases}$$

where $P(\text{promoted}) = \text{logistic}(\beta_0 + \beta_1 j_t^{performance} + \beta_2 \tau_t + \beta_3 j_t^{level})$

Note: Promotion depends on **performance** (function of energy, stress, time at job), not chosen effort.

Market Transitions: 3-state Markov chain with persistence

Health Transitions:

$$Q_{t+1} = f(\text{noise}(L_t), \Sigma_t, \Theta_t, W_t^{\text{intensity}}) + \epsilon_t^Q$$

Reward Function $R_2(s_t, a_t)$

$$R_2(s_t, a_t) = \alpha_1 U^{\text{training}}(m_t^{\text{snatch}}, m_t^{C\&J}, V_t, \Phi_t) + \alpha_2 U^{\text{consumption}}(a_t^{\text{discretionary}}) \\ + \alpha_3 U^{\text{housing}}(L_t) + \alpha_4 U^{\text{security}}(W_t, E_t, I_t) + \alpha_5 U^{\text{health}}(\Phi_t, \Psi_t, Q_t) - \alpha_6 \Sigma_t^2$$

Where:

- $U^{\text{training}} = 5 \cdot (m_t^{\text{snatch}} + m_t^{C\&J})/200 + 3 \cdot (\Phi_t/100) \cdot (V_t/500) + 2 \cdot (1 - \Theta_t/100)$
(combines strength gains and training quality, high weight $\alpha_1 = 8$)
- $U^{\text{consumption}} = 10 \cdot \log(1 + a_t^{\text{discretionary}}/100)$
(log utility from discretionary spending, $\alpha_2 = 2$)
- $U^{\text{housing}} = \text{quality_index}(L_t) \in [0, 100]$
(neighborhood amenities/nightlife/walkability mapped from discrete zone, $\alpha_3 = 3$)
- $U^{\text{security}} = 15 \cdot \sqrt{W_t/10000} + 10 \cdot E_t + 5 \cdot \sqrt{I_t/10000}$
(CRRA utility from wealth and emergency fund, $\alpha_4 = 2.5$)
- $U^{\text{health}} = 0.35 \cdot \Phi_t + 0.35 \cdot \Psi_t + 0.30 \cdot Q_t$
(physical/mental energy and sleep quality, $\alpha_5 = 2$)
- Stress disutility: $\alpha_6 \Sigma_t^2/10000$ (quadratic, $\alpha_6 = 3$)

Discount Factor

$\gamma = 0.998$ (biweekly discounting over 3-year horizon: 78 time steps)

5. Simplified DP Version (Phase 2)

MDP Specification: $\mathcal{M}_3 = (S_3, A_3, P_3, R_3, \gamma)$

Goal: Tractable with exact or approximate dynamic programming before learning RL.

State Space S_3

Fully discrete, 9-dimensional state space:

$$s_t = (W_t, E_t, L_t, W_t^{\text{intensity}}, \Phi_t, M_t, S_t, M_t^{\text{market}}, \tau_t)$$

- $W_t \in \{0, 1, \dots, 10\}$: Wealth level (units of $10k$, $max 100k$)
- $E_t \in \{0, 1, 2\}$: Emergency fund (0=none, 1=partial <3mo expenses, 2=full $\geq 6mo$)
- $L_t \in \{0, 1, 2\}$: Neighborhood tier (0=cheap/far, 1=medium, 2=expensive/close)
- $W_t^{\text{intensity}} \in \{0, 1, 2\}$: Work intensity this month (low/medium/high, stochastic)
- $\Phi_t \in \{0, 1, 2\}$: Energy level (0=low, 1=medium, 2=high)
- $M_t \in \{0, 1, 2\}$: Strength level (0=beginner, 1=intermediate, 2=advanced)
- $S_t \in \{0, 1, 2\}$: Salary tier (0=60k, 1=80k, 2=\$100k+)

- $M_t^{market} \in \{0, 1\}$: Market regime (0=bear, 1=bull)
- $\tau_t \in \{0, 1, \dots, 35\}$: Month (3-year horizon)

State space size: $|S_3| = 11 \times 3 \times 3 \times 3 \times 3 \times 3 \times 2 \times 36 \approx 285,000$ states

Action Space A_3

Fully discrete, 4-dimensional action space:

$$a_t = (a_t^{housing}, a_t^{invest}, a_t^{training}, a_t^{consumption})$$

- $a_t^{housing} \in \{-1, 0, 1\}$: Housing decision (downgrade/stay/upgrade neighborhood)
- $a_t^{invest} \in \{0, 1, 2\}$: Investment level (0%, 10%, 20% of income)
- $a_t^{training} \in \{0, 1, 2\}$: Training intensity (light, moderate, intense)
- $a_t^{consumption} \in \{0, 1, 2\}$: Discretionary spending (frugal, moderate, luxury)

Action space size: $|A_3| = 3 \times 3 \times 3 \times 3 = 81$ actions per state

Transition Dynamics $P_3(s_{t+1}|s_t, a_t)$

Deterministic transitions:

- $W_{t+1} = \text{clip}(W_t + \Delta W_t, 0, 10)$ where $\Delta W_t = f(\text{salary}, \text{rent}(L_t), a_t^{invest}, a_t^{consumption}, \text{market return})$
- $L_{t+1} = \text{clip}(L_t + a_t^{housing}, 0, 2)$
- $\tau_{t+1} = \tau_t + 1$

Stochastic transitions (simple multinomial):

- **Work Intensity** (exogenous, stochastic):
 - $P(W_{t+1}^{intensity}|W_t^{intensity})$ follows mean-reverting process
 - Example: $P(W_{t+1}^{intensity} = 1|W_t^{intensity} = 1) = 0.6$ (moderate persists)
- **Energy:** $P(\Phi_{t+1}|\Phi_t, W_t^{intensity}, a_t^{training}, L_t)$
 - High work intensity + high training $\rightarrow$ likely low energy next period
 - Expensive neighborhood (short commute) $\rightarrow$ energy bonus
 - Example: $P(\Phi_{t+1} = 0|\Phi_t = 2, W_t^{intensity} = 2, a_t^{training} = 2, L_t = 0) = 0.7$
- **Strength:** $P(M_{t+1}|M_t, a_t^{training}, \Phi_t)$
 - Higher training + higher energy $\rightarrow$ increased probability of strength gain
 - Example: $P(M_{t+1} = M_t + 1|M_t = 1, a_t^{training} = 2, \Phi_t = 2) = 0.3$
 - Constraint: $M_{t+1} \in \{\max(0, M_t - 1), M_t, \min(2, M_t + 1)\}$
- **Salary:** $P(S_{t+1}|S_t, W_t^{intensity}, \tau_t)$
 - Promotion probability increases with tenure; high average work intensity helps but is not chosen
 - Example: $P(S_{t+1} = S_t + 1|S_t = 0, \tau \geq 24) = 0.2$ (promotion after 2+ years)

- **Market:** Two-state Markov chain
 - $P(M_{t+1}^{market} = 1 | M_t^{market} = 1) = 0.8$ (bull persists)
 - $P(M_{t+1}^{market} = 0 | M_t^{market} = 0) = 0.6$ (bear persists)
- **Emergency Fund:** Deterministic rule-based (auto-fill to target before investing)

Reward Function $R_3(s_t, a_t)$

$$R_3(s_t, a_t) = \alpha_1 M_t + \alpha_2 \Phi_t + \alpha_3 L_t + \alpha_4 \sqrt{W_t} + \alpha_5 a_t^{consumption} - \alpha_6 \cdot 1_{E_t < 2} \cdot 1_{W_t < 3}$$

Where:

- $\alpha_1 = 5$: Strength utility (training performance is primary goal)
- $\alpha_2 = 2$: Energy utility (feeling energized)
- $\alpha_3 = 1.5$: Neighborhood quality (cafes, nightlife, walkability)
- $\alpha_4 = 2$: Wealth security (diminishing returns via $\sqrt{\cdot}$)
- $\alpha_5 = 1$: Consumption enjoyment
- $\alpha_6 = 10$: Stress penalty (low wealth + no emergency fund)

Terminal reward: $R_T(s_T) = 10 \cdot W_T + 5 \cdot M_T$ (final wealth and strength)

Discount Factor

$\gamma = 0.99$ (monthly discounting over 3-year horizon: 36 time steps)

Solution Method

- **Exact:** Backward value iteration with full state enumeration
- **Approximate:** Linear function approximation if state space too large

6. Code Implementation

The MDP is implemented as a parameterized Python class `MultiCapitalEnergyMDP` that can instantiate all three versions through configuration parameters. See separate file `multi_capital_mdp.py` for full implementation.

Class Structure:

- Extends `MarkovDecisionProcess[S, A]` base class (from course materials)
- `MultiCapitalEnergyMDP(version='v1'/'v2'/'v3')` instantiates the desired version
- Currently implements V3 (simplified discrete version for Phase 2)
- V2 and V1 will extend this structure with continuous state spaces

State and Action Spaces:

- `StateV3` : Fully discrete 9-dimensional dataclass (frozen for hashing)
 - Includes: wealth, emergency, location, work_intensity, energy, strength, salary, market, time

- `ActionV3` : Fully discrete 4-dimensional dataclass
 - Includes: housing, invest, training, consumption
- Dataclasses provide type safety and enable efficient state enumeration for DP

Transition Dynamics:

- `step(state, action)` returns `Distribution[Tuple[State, float]]`
- Implements sampling-based transitions with stochastic components:
 - Work intensity: Exogenous AR(1) process
 - Energy: Depends on work, training, and location (commute time proxy)
 - Strength: Depends on training intensity and available energy
 - Salary: Promotion based on tenure and work intensity
 - Market: Two-state persistent Markov chain
- Deterministic components: wealth accumulation, location choice, time

Reward Function:

- Multi-objective utility with 6 components
- Parameterized by weights α_i (can be tuned)
- Includes terminal reward for final wealth and strength

State Space Size:

- V3: $11 \times 3 \times 3 \times 3 \times 3 \times 3 \times 2 \times 36 \approx 577,000$ states
- Tractable with exact backward value iteration (Phase 2)
- Action space: $3 \times 3 \times 3 \times 3 = 81$ actions per state

Example Usage

```
from multi_capital_mdp import MultiCapitalEnergyMDP, StateV3, NonTerminal

# Create MDP instance
mdp = MultiCapitalEnergyMDP(version='v3', gamma=0.99, time_horizon=36)

# Create initial state
state = NonTerminal(StateV3(
    wealth=5, emergency=1, location=1, work_intensity=1,
    energy=2, strength=0, salary=0, market=1, time=0
))

# Get available actions and sample transition
actions = list(mdp.actions(state)) # 81 actions available
next_dist = mdp.step(state, actions[0])
next_state, reward = next_dist.sample()
```

The implementation demonstrates the curse of dimensionality (577k states for simplified version) and motivates the need for function approximation in V2 and V1.